

Interim Summary

What Was Built

A two-notebook analytical pipeline linking **financial news sentiment** to **stock price movements**, covering Tasks 1 and 2 of the Nova Financial Solutions challenge.

Task 1 — Exploratory Data Analysis

Dataset: 800 news articles · 3 tickers (AAPL, MSFT, GOOGL) · Jan 2023 onwards

Finding	Detail
Headline length	Mean 138 chars / 19 words — uniform, wire-service style
Top publishers	benzinga.com (17%), reuters.com (16%), thefly.com (16%), marketwatch.com (15%) — top 4 cover 64% of articles
Publication timing	Bimodal: peaks at 07:00–09:00 ET (pre-market) and 15:00–16:00 ET (close)
Zero data loss	0% unparseable timestamps after stripping UTC-4 suffix

4 LDA topics identified in headlines:

- 1. **Biotech catalyst** — FDA approval, pipeline, stock jumps
- 2. **Earnings surprise** — beat estimates, shares rise, downgrade
- 3. **Analyst upgrade** — price target raised, bullish
- 4. **Income / yield** — dividend increased, yield, income

These themes map directly to known market microstructure patterns and will serve as categorical features in Task 3 correlation modelling.

Task 2 — Technical Indicators (TA-Lib)

Dataset: AAPL · 320 trading days · Price range \$280 – \$422

Indicator	Parameters	Last Reading (25 Mar 2024)
SMA	20-day, 50-day	\$381.77 / \$370.66 — price well above both
EMA	20-day	\$389.00 — faster trend tracking
RSI	14-day	67.3 — pulling back from overbought (peaked 75.2)
MACD	12/26/9	+13.71 — positive momentum, no reversal signal yet

Key signal: Price is in a strong uptrend with RSI recently overbought and MACD histogram expanding — momentum is intact but a short-term correction is plausible.

Plan for Final Submission (12 May 2026)

Task	Action
Task 1 (full data)	Swap sample CSVs for full FNSPID slice; annotate publication spikes with FOMC/CPI dates
Task 2 (expand)	Add Bollinger Bands and ATR
Task 3 (new)	VADER sentiment scores → date alignment → Pearson correlation with daily returns → investment strategy recommendations

GitHub: [rediet-shewarega/predicting-price-moves-news-sentiment-](https://github.com/rediet-shewarega/predicting-price-moves-news-sentiment-) · Branch: `task-1`